City and Environment Interactions 31 (2026) 100423

Contents lists available at ScienceDirect

City and Environment Interactions

journal homepage: www.sciencedirect.com/journal/city-and-environment-interactions

GIS-based prioritization of urban green roofs to enhance biodiversity in Amman, Jordan

Razan Alhmoud [a,1,*], Andreas Braun [b,2], Katja Tielbörger [a,3,4]

[a] Institute of Evolution and Ecology, Plant Ecology, University of Tübingen, 72076 Tübingen, Germany
[b] Department of Geosciences, Geography, University of Tübingen, 72076 Tübingen, Germany

ARTICLE INFO

Keywords:
Urban biodiversity
Green roofs
GIS
Multi-criteria analysis
Urban planning
Semi-arid cities
Decision-support

ABSTRACT

Green roofs (GR) can support biodiversity and improve microclimatic conditions, yet city-scale GR planning in semi-arid and economically poorer regions remains limited. We developed a GIS-based multi-criteria approach to identify priority zones for GR implementation in Amman, Jordan, considering ecological, climatic, and socio-economic factors such as proximity to existing greenery, favourable climate (higher precipitation, lower temperatures), and affordability. We compared a context-specific semi-arid approach with a conventional temperate-region approach, as well as four scenarios with different relative weights of each criterion. Our results showed that semi-arid priority zones were concentrated in cooler, wetter areas suitable for GR establishment, while temperate-prioritized zones overlapped with hotter, drier locations where GR feasibility may be limited but the need for GR may be higher. We also found low-regret areas that emerged as high-priority zones irrespective of the ecological and economic weightings. Our findings highlight the importance of context-specific GR planning and provide a practical GIS-based decision-support tool that could be beneficial for urban biodiversity enhancement in arid cities of the Global South. This GIS-based framework, which stresses the *feasibility* of GR implementation, can guide policymakers and urban planners in prioritizing areas for GR implementation that may maximize biodiversity and climate adaptation benefits in water-scarce cities.

1. Introduction

Urbanization contributes to green space loss, threatening biodiversity and accelerating the "sixth mass extinction" [1,2]. Mitigating these effects requires increasing the quantity, quality, and accessibility of green spaces through efficient planning [3–5]. Green Roofs (GR) function as multifunctional and nature-based solutions that provide essential ecosystem services (e.g., water retention, microclimate improvement) and may enhance urban biodiversity and have recently become increasingly promoted [6–8]. Yet, most GR research has been conducted in the Global North, while cities in the Global South -where environmental and socio-economic constraints are often much more pronounced - remain scarce. This is regrettable, as promotion of urban green space is likely to be particularly important in regions with harsh environmental conditions and a faster rate of urbanization. Therefore, methods for GR implementation that are applicable across a range of ecological and socio-economic conditions are dearly needed.

GR planning should be conducted at the city scale, as their function for promoting biodiversity is strongly influenced by their spatial context. For example, island biogeography theory suggests that larger islands and islands closer to the mainland (or other islands) host more species than small, isolated islands [9]. Therefore, GR with a larger area, as well as GR situated closer to existing natural greenery, should support more species [7,10,11], and strategically distributed GR can thus create habitat networks for the movement of individuals of species, enabling rescue of populations that may go extinct on isolated roofs [1,12]. These

* Corresponding author.
E-mail addresses: razan-jabir-nawaf.al-hmoud@student.uni-tuebingen.de (R. Alhmoud), an.braun@uni-tuebingen.de (A. Braun), katja.tielboerger@uni-tuebingen.de (K. Tielbörger).
[1] University of Tübingen, Institute of Evolution and Ecology, Plant Ecology Group, Auf der Morgenstelle 5, D-72076 Tübingen.
[2] Universität Tübingen, Department of Geosciences, Geographical Institute, Rümelinstr. 19–23, D-72070 Tübingen.
[3] University of Tübingen, Institute of Evolution and Ecology, Plant Ecology Group, Auf der Morgenstelle 5, D-72076 Tübingen.
[4] Cluster of Excellence (EXC 3121): TERRA – Terrestrial Geo-Biosphere Interactions in a Changing World, University of Tübingen, Germany. Cluster of Excellence GreenRobust, Eberhard Karls University, Tübingen 72076, Germany.

https://doi.org/10.1016/j.cacint.2026.100423
Received 16 February 2026; Received in revised form 18 June 2026; Accepted 26 June 2026
Available online 1 July 2026
2590-2520/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

studies point to the importance of planning at the city scale, in addition to the manifold studies examining individual GR designs [13,14].

Several previous studies have adopted such a city-scale approach. However, these have primarily been conducted in the Global North, and more often in temperate regions, where establishment of GR may not be limited by water availability or socio-economic constraints. Also, even when studies have focused on drier regions or cities of the Global South, the commonly used approach, which we call the 'temperate' approach in this study, has been adopted for prioritizing areas for GR establishment. This frequently used scheme in that context would, for example, prioritize areas with a larger urban heat island effect and (correlated to the former) low biodiversity and low percentage of – or larger distance to – green space [e.g.10, 15, 16, 17]. The rationale for this prioritization is that GR should be established where they are most *needed*, i.e. where implementation could have the largest beneficial effect for (micro) climate and/or biodiversity [16]. However, for several reasons, this approach may yield highly unsatisfying and unrealistic results when robotically applied outside the temperate North.

First, roofs are challenging habitats for establishing vegetation, as they usually allow only for relatively shallow substrate, and they are exposed to more extreme temperatures and wind [18]. Therefore, vegetation on GR is usually water-limited, even in temperate regions [19], resulting in a rather small number of drought-tolerant species used for greening roofs [20]. This problem is aggravated in climates characterized by low precipitation and high temperatures, such as Mediterranean to arid climates, where summers with no rain and droughts during the rainy season may strongly limit plant growth. Unfortunately, studies about urban greening in such climates are still relatively rare. A few studies have demonstrated that implementing GR in Mediterranean areas, albeit with relatively low temperatures and high precipitation, such as in Cosenza and Grosseto in Italy [21,22] and Bilbao in Spain [23], is beneficial for vegetation persistence. Interestingly, some of the few studies in water-limited climates have adopted the above 'temperate' approach and prioritized areas with higher temperatures in Mediterranean cities like Seville and Barcelona in Spain [15,16].

However, in dry regions, this may be counterproductive, as not only the potential benefits but also the *feasibility* dictate whether GR should be established in a particular place. GR vegetation can survive in dry climates during periods without natural precipitation when designed with drought-tolerant plants, deeper substrate, and efficient water storage systems [24,25]. Nevertheless, under extreme climatic stress, additional irrigation may be necessary for non-succulent species [26–28]. Such irrigation systems, however, can be very expensive [6], imposing not only ecological but also socio-economic constraints on GR establishment.

This points to another understudied aspect for city-scale guidelines for GR establishment. Namely, socio-economic conditions such as affordability are rarely integrated into GR planning, despite their importance. Moreover, joint consideration of economic, physical, and ecological aspects is rare, and when applied, they are again limited to situations of 'temperate' North [7,16]. This creates a bias in existing prioritization schemes because stronger economies may support subsidies [6,29]. Therefore, these studies are again targeting areas where GR may be most *needed*: socio-economically disadvantaged neighbourhoods are prioritized for reducing inequality [7,16].

Here, we argue that this focus is likely to fail in cities where ecological conditions are harsh (i.e., water limitation), and where economic conditions are not suited to buffer these conditions, e.g., via expensive supplemental irrigation technologies. It should also be noted that only a few people are able and willing to invest in GR [7,30], even in regions where subsidies are available. Thus, the likelihood that GR are implemented in arid regions, which are often economically poor, and where governmental incentives and subsidies are absent [31], is rather low and would only be possible for wealthier individuals who can afford the setup as well as the costly (i.e., irrigation-intensive) maintenance. Wealth status is typically approximated by demographic (e.g., household size, density) and economic (e.g., income, unemployment, and literacy) indicators [32]. Accordingly, bigger household size and higher income may be relevant.

Prioritization criteria also vary by goal and context. For example, for mitigating the urban heat island effect (UHI), areas with little green cover and relatively high building density are prioritized [15,17]. If improvement of air quality is the main goal, proximity to roads and low greenery areas matter. Vice versa, in arid regions, areas closer to greenery, more favourable climatic conditions, and higher socio-economic affordability could be the focus of GR implementation, which may enhance urban biodiversity, as this is where both ecological and economic conditions enable the implementation. Overall, a flexible GR planning scheme is needed that can be locally adapted to different climatic, ecological, and socio-economic conditions and that can address either (or both) the *need* for GR or the *feasibility*.

Given that several criteria are involved and that the weighting of them may largely differ among cities and regions, such a flexible approach also enables prioritizing different stakeholders' requirements according to their preferences. Therefore, spatially-explicit Multi-Criteria Decision Approaches (MCDA) have been advocated as an ideal approach for determining priority areas for GR implementation. For example, Joshi et al. (2020) used surface temperature, green cover, and socio-economic criteria for their city-scale prioritization. A challenge in such approaches is then the weighting of these criteria relative to each other. Namely, using criteria that may support both biodiversity (e.g., closer distance to greenery as a proxy) and ecosystem services (e.g., thermal relief in areas with little greenery) may create spatial conflicts. Grunwald et al. (2017) attempted to resolve this by setting thresholds representing a compromise (e.g., 'intermediate' distance to greenery), but such conditions rarely occur, and the values may be set in a subjective manner. Recognizing that not all criteria are equally relevant in all contexts, Langemeyer et al. (2020) incorporated expert-driven criteria weighting assessments to prioritize zones. Yet, Langemeyer et al. (2020) still used only a single weighting scenario, i.e., the sensitivity of differential weightings (e.g., in different settings by different stakeholders) remained unexplored. This is regrettable, as GIS tools such as "weighted overlay" tool, which applies MCDA, provide a relatively easy way of flexibly applying different weighting scenarios.

In this study, the "temperate" approach prioritizes areas with the greatest *need* for GR implementation, including areas with a low percentage of, or greater distance to, greenery, harsh climatic conditions, such as higher surface temperatures, and disadvantaged neighbourhoods. In contrast, our "semi-arid" approach prioritizes areas with the greatest *feasibility* for GR implementation, which may also enhance urban biodiversity, including closer proximity to greenery, more favourable climatic conditions (higher precipitation and lower maximum temperatures), and higher socio-economic affordability indicators (such as higher income, lower unemployment, and larger household size).

We primarily focus on the results of a locally-adapted "semi-arid" approach but also compare results between approaches, which essentially represent opposite prioritization logics. Here, we use Amman, Jordan, as an ideal case study to address the above research gaps. Namely, we are searching for the most suitable prioritization approach in an area constrained by climatic as well as socio-economic conditions. We are applying the feasibility-focused "semi-arid" approach, which we assume is tailored to Amman's local conditions, using different weighting scenarios. We assume that this locally adapted approach will yield more relevant results than a need-based approach, but also test for low regret strategies, i.e., whether there would be prioritization zones that are irrespective of the approach.

2. Study area, data, and methods

2.1. Study area

Our study area is in Jordan, the second most water-scarce country in the world, where socio-economic conditions are very challenging. Jordan is located in a global biodiversity hotspot [33,34] and thus bears special responsibility for biodiversity conservation. The population growth is large (10,5 million in 2019) and has recently greatly accelerated due to refugee influxes. This has contributed to an 83% rise in public debt, with 14.4% of the population living in poverty, especially in Amman, where low incomes meet high living costs, with an average monthly household income of ~818 USD, and ~ 409 USD for refugees, respectively [35].

The Greater Amman Municipality (GAM) administrative borders (Fig. 1) cover an area of ~800 km^2. Amman has experienced rapid urban expansion, with about 4 million residents in 2015, which is projected to reach 6.4 million by 2025 [35]. The green space is extremely limited, with only 2.5 m^2 per capita, i.e., significantly below the international benchmark of at least 10 m^2. Overall, Amman has lost nearly half of its vegetation cover over the past 30 years, mainly due to urbanization [34–37], which has negatively impacted biodiversity [34]. Yet, detailed data about biodiversity is missing in Amman. Amman exhibits a steep climatic gradient, ranging from a semiarid climate (18 °C MAT, 450 mm AAP) in the northwest to arid in the southeast (30 °C MAT, 150 mm AAP) [32,38]. Predominantly flat roofs in the region offer suitable structural conditions for GR installation. Although interest in GR is growing, as emphasized in the Green City Action Plan 2021, which promotes expanding green spaces through parks, gardens, and micro-scale infrastructure such as GR [34], they remain rare. The number, location, type, and condition of GR in Amman are largely undocumented, with existing installations likely numbering only a few dozen to a couple of hundred, implemented privately or by NGOs such as Greening the Camps and Durrat Almanal, which support urban greening [39,40]. Installation costs are high: a GR unit of only ~10 m^2 costs ~3525 USD excluding labour [41], exceeding four times the average monthly household income and more than eight times that of a refugee household.

2.2. Map datasets

We used several map datasets with the GAM borders delineating the study area. Five maps were employed to produce priority zones: one land cover map, two climatic maps, and two socio-economic affordability maps. Details of all maps are listed in Table 1 in Appendix A. Additionally, a map of all city roofs was used to select larger suitable ones, which were prioritized within each zone and across all scenarios, as in most previous studies [7,10,11,17]. Since our primary objective was to prioritize areas and not a single roof, findings on final roof suitability are presented in Appendix B.

Maps were obtained mainly from open-access platforms and governmental reports. The GAM boundary map was extracted as a JPEG from the 'Amman Spatial Profile' governmental report [35]. Land cover maps were downloaded from the European Space Agency [42] as a raster with 10 m resolution and converted to a polygon shapefile. Climatic and socio-economic maps were extracted as JPEGs from the 'Jordanian Fourth National Communication' governmental report [JFNC report; 32]. Roof maps were downloaded from GitHub [43] as polygon shapefiles. All maps were processed using ArcGIS Pro 3.0.0 [44]. GAM boundary, climatic, and socio-economic maps were originally in JPEG format and were rescaled and manually vectorized into polygon shapefiles. Layers were projected to a common coordinate system and clipped to the extent of the GAM boundary.

We acknowledge the methodological limitations associated with the use of spatial layers derived from JPEG images, which were necessary due to the limited availability of spatial data in the region. However, a

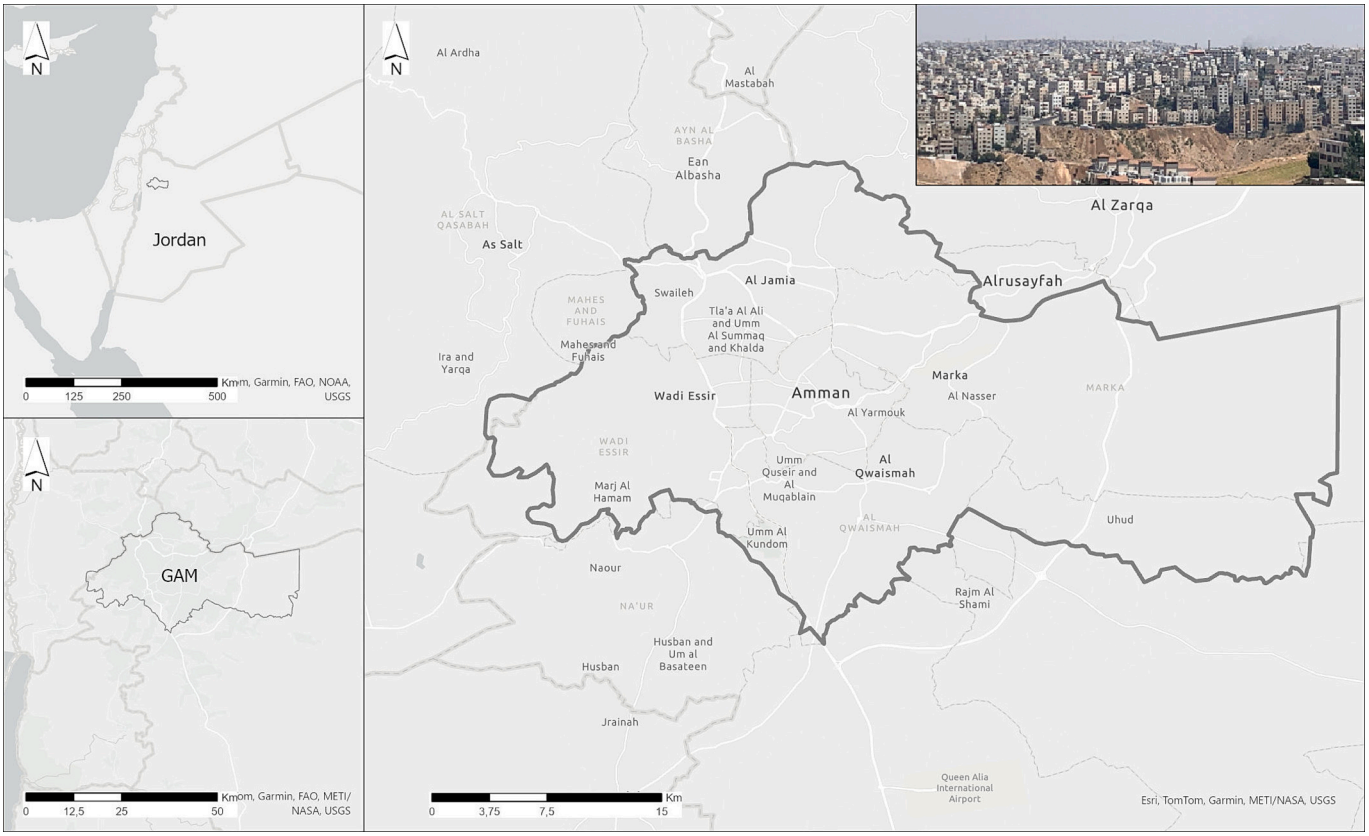


Fig. 1. The study area GAM borders [35], Amman city from Taj mall (Photographer: Author, 2022).

structured workflow was implemented to minimize positional and classification errors. All images were georeferenced using stable reference points and a first-order polynomial (affine) transformation in WGS 84, with accuracy assessed using RMSE based on 9–30 control points per layer. Manual digitization followed the original map legends. Overall, while some degree of uncertainty and bias may remain, we believe that this does not change the overall conclusions of our findings. Namely, the different prioritization areas are very large (several km^2), compared to the estimated spatial uncertainty ranging from hundreds of metres to a few kilometres (see Appendix C for details).

2.3. Criteria for priority zones

The land cover map was used to assess the distance to existing greenery, assuming that proximity facilitates species movement. Climatic maps included (1) Mean Annual Precipitation (MAP) and (2) Maximum Air Temperature (MAT) and were used to identify areas with climatic conditions that are more favourable for plant establishment on

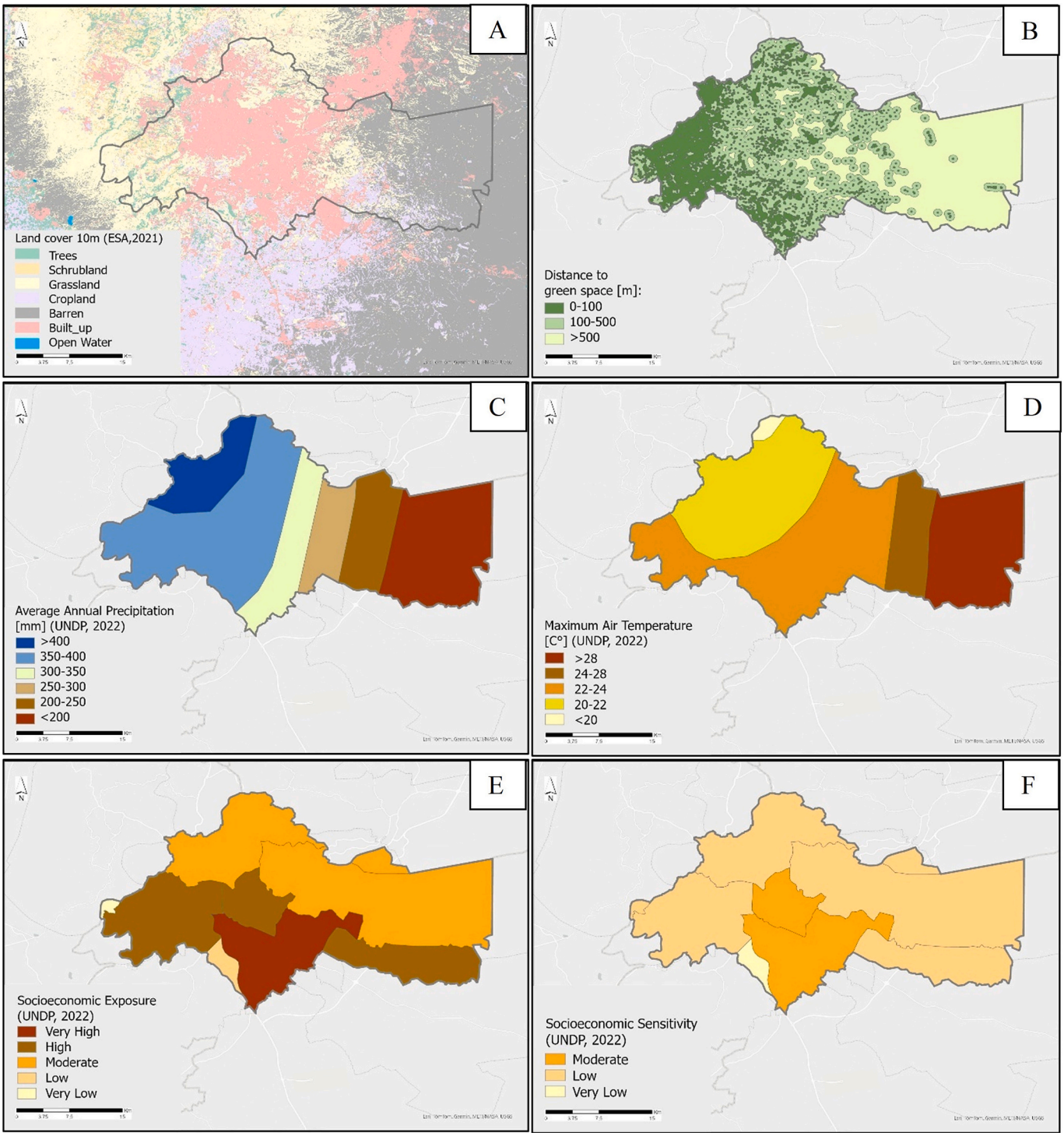


Fig. 2. Baseline maps for the study area (Greater Amman Municipality): A - Land Cover, B- Distance to green space (trees & shrubs), C- Average Annual Precipitation. D- Maximum Air Temperature, *E*- Socio-economic Exposure Map, F- Socio-economic Sensitivity Map (see main text for data sources). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

GR (higher precipitation and lower temperature). Socio-economic affordability maps included (1) Exposure maps (Demographic indicators), and (2) Sensitivity maps (Economic indicators), and were used to identify districts with higher affordability for GR implementation (higher Demographic and Economic indicators), while the JFNC report applied a reverse approach to assess vulnerability (Fig. 2 A-F). In the original report, these maps were combined to assess socio-economic vulnerability, such that areas with higher demographic and economic capacity were assigned lower vulnerability. In our study, we used the maps directly, without reversing their underlying values, and interpreted higher values as indicating greater socio-economic affordability and capacity to support GR implementation. Importantly, no recalculation or modification of the original layers was performed; only their interpretation was adapted to align with districts with potentially greater feasibility for GR adoption.

2.4. Original classifications

Land Cover: We extracted the *Tree* and *Shrubland* classes from the land cover data (Fig. 2A) to represent existing natural greenery. We then created areas classified by their proximity to existing natural greenery at distances of 0–100 m, 100–500 m, and > 500 m around these classes, respectively, following Grunwald et al. (2017) by using the buffer tool in ArcGIS Pro (Fig. 2 B). Based on previous studies, we assumed that GR located closer to natural green areas should enhance ecological connectivity and thus may support higher biodiversity [8,45]. This map resulted in a noticeable gradient, with denser greenery concentrated in the western parts of the city, while the eastern regions represented predominantly barren land (Fig. 2B).

Climate maps: In our climatic maps, 1) Mean Annual Precipitation was originally categorized into areas divided by 50 mm MAP, i.e.,>400, 350–400, 300–350, 250–300, 200–250, and < 200 mm, respectively (Fig. 2C). 2) Maximum Air Temperature was originally categorized in increments of 2 °C as: >28 °C, 24–28 °C, 22–24 °C, 20–22 °C, and < 20 °C (Fig. 2D). A noticeable climatic gradient could also be observed with a more benign climate with higher precipitation and lower temperatures toward the northwest and drier, hotter conditions in the southeast.

Socio-economic affordability maps: In our maps, 1) the Exposure map originally included five categories: Very High, High, Moderate, Low, and Very Low, respectively (Fig. 2E), while the 2) Sensitivity map originally included the three categories Moderate, Low, and Very Low, respectively (Fig. 2F). A noticeable socio-economic affordability gradient was observed, with wealthier individuals predominantly concentrated in the south-western parts of the city.

2.5. Normalizing classes

Following Joshi et al. (2020), varied classes were normalized into numeric ranks with comparable ranges to ensure consistency across criteria, as summarized in Table 1. Ranks were adjusted to the Amman-specific conditions (i.e., 'Semi-arid' approach), which ranged from 1 (high) to 6 (low). Climatic and socio-economic maps retained the original class intervals and categories from the source maps. To enable integration of variables with different numbers of classes, each class was converted into an ordinal suitability rank that preserves its relative position along the underlying gradient. This approach was used because our analysis is based on relative ordering rather than absolute differences between classes, allowing variables with different categorical resolutions to be combined consistently in a weighted overlay framework [46].

2.6. Priority zones

We used the "weighted overlay" tool in ArcGIS Pro to identify the priority zones. In a first step, we integrated the maps from the previous sections (Fig. 2 B—F) by assigning equal weights to all criteria. The resulting map indicates three Priority Zones (Fig. 3), which are easy to interpret. Zone 1 includes high-priority areas, Zone 2 moderate-priority areas, and Zone 3 low-priority areas. It should be noted that changing the rank, scale, or weight can change the results of the overlay outcome. Weight and scale within the tool can be manually adjusted. We used a scale of 1–5.

In a second step, we varied the weight of individual criteria and investigated the effect of this weighting on prioritization outcomes. Specifically, we created four scenarios (I–IV, Table 2). Scenario I (equal weights) served as a neutral baseline. Scenario II emphasized proximity to greenery and favourable climate. Scenario III emphasized proximity to greenery and higher socio-economic affordability. Scenario IV, in contrast, emphasized higher socio-economic affordability while assigning lower importance to ecological criteria. Our main goal here was to test to what extent different weighting strategies influence the spatial prioritization of GR implementation, but not how exact weights play out. I.e., we aimed to demonstrate how weighting of either ecological or economic criteria influences prioritization outcomes, while leaving the selection of weights to decision-makers.

Therefore, we selected a limited number of scenarios which differed substantially in the weighting of key factors while maintaining "realistic" weights for more objective variables. As a baseline, we used an "all equal-scenario" as in Joshi (2020). We expanded this baseline on the one hand to a scenario with high weight on ecological parameters (here: proximity to greenery as a strong predictor of GR success [8,45]) and, on the other hand, to a stronger focus on the Socio-economic layers (scenario IV). Subjective criteria were weighted in a more heuristic manner as a proof-of-principle to illustrate how both weighting assumptions and resulting outcomes vary. Multiple weighting scenarios were applied to reduce potential subjectivity in weight assignment, as suggested by Langemeyer et al. (2020), who applied a single stakeholder-informed scenario. All weights were constrained to sum to 100% as required by the GIS tool. The analysis included five input layers, with the two climatic and two socio-economic variables treated relatively evenly across scenarios to avoid overemphasizing any single climatic or socio-economic dimension.

The weighted overlay tool uses a raster-based multi-criteria decision analysis. All inputs from the five maps were first converted to a raster format using the default ArcGIS Pro vector-to-raster conversion settings

Table 1
Normalized ranking of varied classes across criteria using the 'Semi-arid' approach.

Rank	Proximity to green space	Climatic Conditions		Socio-economic affordability	
	Greenery distance [m]	Average Annual Precipitation [mm]	Maximum Air Temperature [°C]	(Demographic) exposure [V.High-V.Low]	(Economic) sensitivity [Moderate-V. Low]
1	0–100	400–450	<20	Very High	
2	100–500	350–400	20–22	High	
3	>500	300–350	22–24	Moderate	Moderate
4		250–300	24–28	Low	Low
5		200–250	>28	Very Low	Very Low
6		<200			

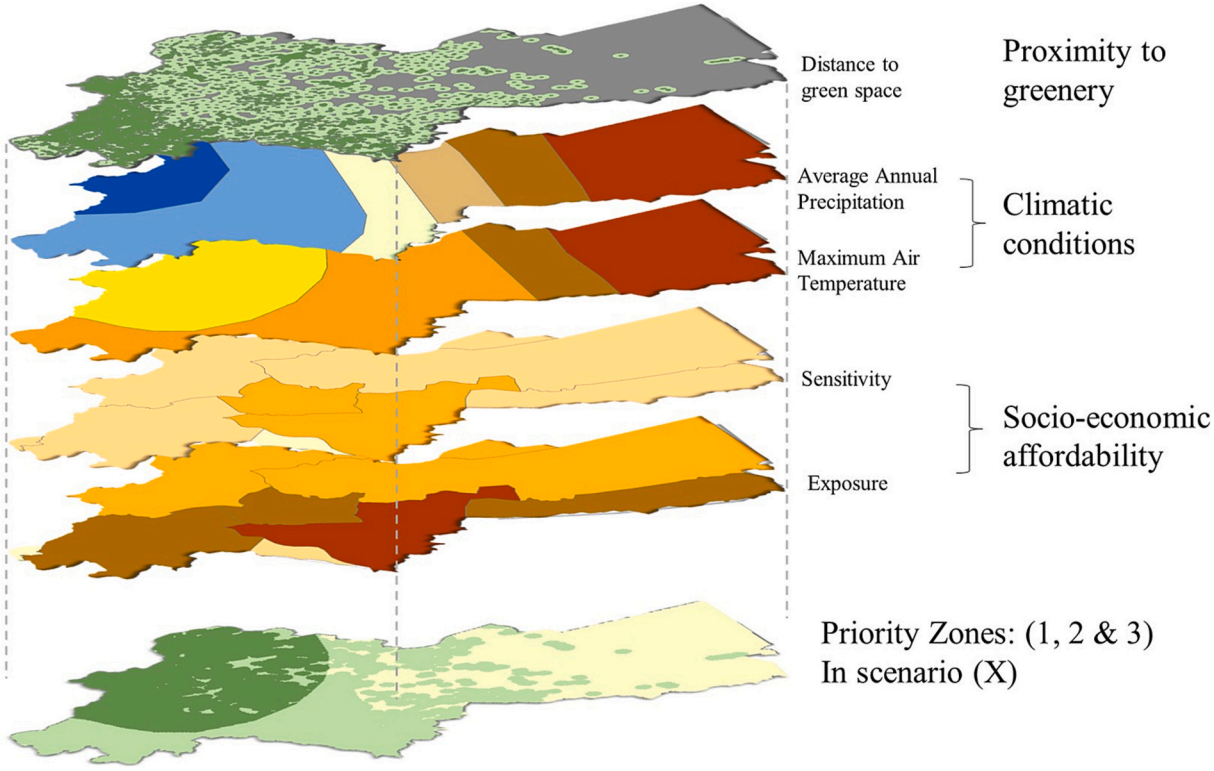


Fig. 3. Schematics illustrating the usage of the Multi-Criteria Decision-making Approach (MCDA) for prioritizing zones.

Table 2
Weighting scenarios I-IV.

Scenario Nr.	Scenario focus	Weight Proximity to Greenery	Weight Climatic		Weight Socio-economic affordability	
			(Temperature)	(Precipitation)	(Demographic) Exposure	(Economic) Sensitivity
Scenario I)	All Equal	20	20	20	20	20
Scenario II)	Ecological (Greenery) with Climatic Focus:	40	20	20	10	10
Scenario III)	Ecological (Greenery) with Socio-economic Focus:	40	10	10	20	20
Scenario IV)	Socio-economic Focus:	20	10	10	30	30

with a uniform cell size of approximately 0.00011° to ensure spatial consistency across datasets. The nearest-neighbour resampling method was applied to preserve original class values without introducing interpolation artefacts, which is appropriate for categorical suitability data. Each raster contained ordinal rank values (1–6, where 1 indicates the highest suitability) as assigned during the normalization step (Table 1). Within the tool's reclassification table (WOTable), these rank values were mapped to the ArcGIS internal evaluation scale of 1–5, with rank 5/6 pooled to a scale value of 5 (lowest suitability). The tool then combined the five raster maps using scenario-specific weights (Scenarios I–IV), producing a suitability raster with values of 1, 2, and 3, respectively, representing a semi-continuous output based on a weighted linear combination of input layers as in the formula below. The resulting output was subsequently classified into three final suitability zones (1 = high suitability, 2 = moderate suitability, and 3 = low suitability).

The ArcGIS Pro Weighted Overlay formula:

$$\text{Output Value} = \text{Round}\left(\sum_{i=1}^{n}(R_i \times W_i)\right)$$

where:

- n is the number of input raster layers (five in this study),
- R_i is the standardized rank assigned to raster i,
- W_i is the percentage weight assigned to raster i (expressed as a decimal),
- Round indicates rounding to the nearest integer, as the output raster contains integer values.

We then visually compared the spatial patterns of GR priority zones for the two opposing approaches (semi-arid vs. temperate) and for the different scenarios. Further, we extracted high-priority areas (Zone 1) from all four weighting scenarios of the 'Semi-arid' approach as polygons and overlaid them in a single map. Areas that overlapped spatially were considered stable and insensitive to weighting changes, whereas non-overlapping areas that shifted in location across scenarios were considered sensitive.

3. Results

3.1. Priority zones using the 'semi-arid' approach across scenarios I-IV

In (Fig. 4 A-D), all scenarios showed a tendency for prioritizing central to western areas as exhibiting potentially higher *feasibility* for GR implementation.

However, the different weightings resulted in some shifts in the

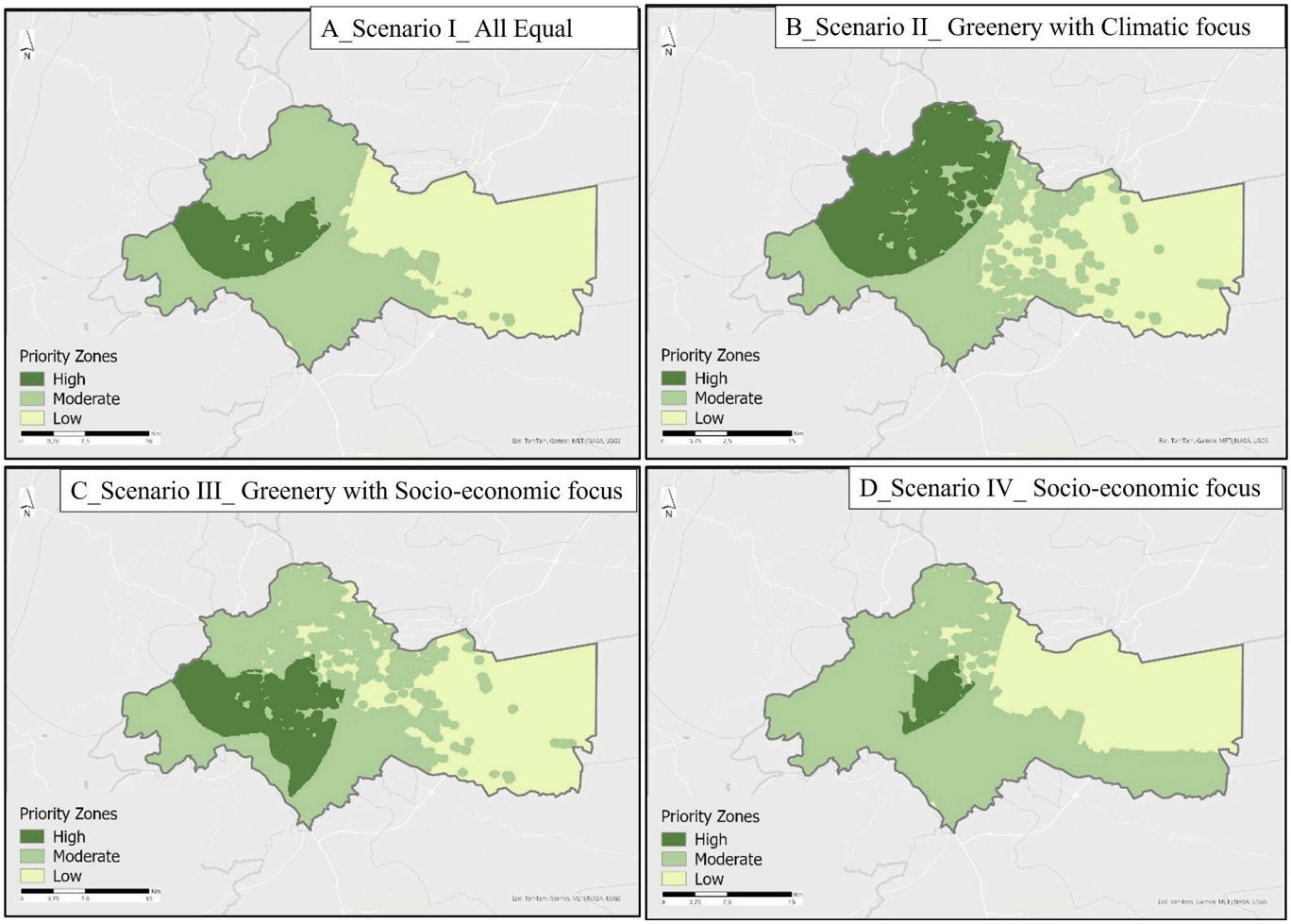


Fig. 4. A-D Priority zones 'Semi-arid' approach across scenarios I–IV.

priority zones. Namely, climate-driven prioritization earmarked a relatively large fraction of the city in the North-West (Fig. 4b), while adding socio-economic factors in the other three scenarios led to shrinkage of the Northern sections, down to a very small priority zone when socio-economic factors were weighted strongly (Fig. 4d). Equal prioritization produced a balanced distribution from the centre toward the west (Fig. 4a). The location of low-priority zones (3) was consistently found in the East. The relative area of the high-priority zone also differed largely among the scenarios, with 11.2%, 27.5%, 16.9%, and 3.8% of the area in Scenarios I–IV, respectively.

3.2. *'Semi-arid' vs. 'temperate' approach (feasibility vs. need)*

Applying the 'temperate' approach, which emphasizes *need* rather than *feasibility*, produced a mirror-image pattern across all scenarios (e.g., Fig. 5). Namely, the highest priority area shifted to the eastern part of GAM, which consists largely of barren land and is located farthest from the main greenery in the west, within ~40 km away. Distances to small green patches in this area are mostly greater than 500 m. This area is also the driest and hottest part of GAM (~219 mm [AAP], ~27 °C [MAT]) and is characterized by lower affordability and lower urban density. In contrast, applying the 'semi-arid' approach assigned the highest priority to the mid-western part of GAM, where larger trees and shrubs are more abundant and where the main greenery in the west is located within ~5 km. Distances to small green patches are mostly less than 500 m. This area receives higher rainfall and experiences cooler temperatures (~ 375 mm [AAP] and ~ 21 [MAT]), while also exhibiting higher affordability and greater urban density. These are conditions that may make

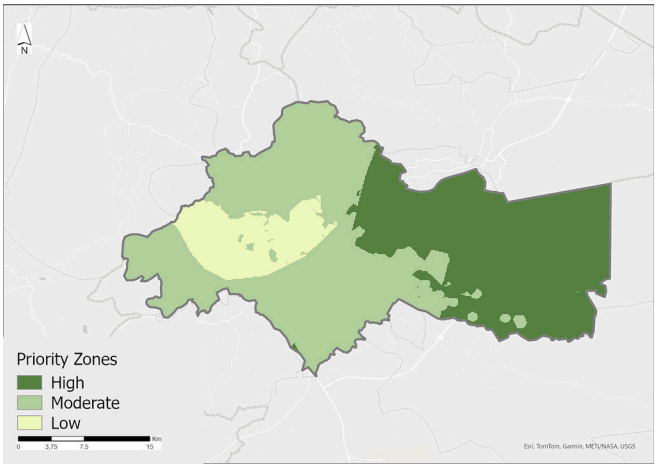


Fig. 5. Corresponding 'Temperate' approach results with inverse spatial patterns for prioritization across scenario (I) as an example.

these areas more suitable and *feasible* for GR implementation, with potential benefits for enhancing biodiversity in Amman.

The "temperate" approach was derived from the dominating assumptions in previous GR studies, which mainly stem from non-arid conditions. It was included to illustrate how this approach, which prioritizes areas with ecologically, climatically, and socio-economically

constrained conditions (i.e. need-based)—performs under semi-arid conditions. While we obviously expect a 'simple' mirror-image of our locally adapted approach to the temperate one, our graphical findings should show visually and clearly how large the differences in the prioritization may actually be.

3.3. Sensitivity of high-priority zones using the 'semi-arid' approach across scenarios

In (Fig. 6), the centre of Amman (Dark Green) consistently emerged as a high-priority zone regardless of the weighting scenario applied, reflecting high spatial stability for GR implementation. Outside this core, western areas along the north–south axis fluctuated between high- and moderate-priority zones depending on weighting, indicating trade-offs between ecological, climatic, and socio-economic criteria. Specifically, north-western areas were particularly sensitive to ecological and climatic weighting (Red), south-western areas to ecological and socio-economic weighting (Black), while the intermediate zone (Bright Green) reflected a region where all criteria were equal. Quantitatively, the stable core (Dark Green) aligns with Scenario IV (100%), Scenario I (Bright Green) has the highest overlap with the core (33.7%), followed by Scenario III (black) with (22.4%) overlap, and Scenario II (red) with (13.7%) overlap.

Jaccard similarity indices (intersection-over-union) were computed to confirm and quantify the spatial divergence observed across weighting scenarios, using Scenario IV as reference. Scenario IV was selected as the reference because it represents the highest spatial agreement (overlap) among scenarios, providing a stable baseline for comparison in the Jaccard analysis. Scenario II, emphasizing climatic suitability, showed the lowest agreement with the socio-economically focused reference scenario ($J = 0.137$), indicating that climatically favourable areas in the northwest and socio-economically affordable areas in the southwest are spatially distinct (less than 1/7 overlap of their combined high-priority area). Scenario III ($J = 0.224$) and Scenario I ($J = 0.337$) show stronger agreement with Scenario IV, consistent with the stepwise shift from climatic toward socio-economic weighting. Together, these values indicate that the spatial configuration of high-priority zones is highly sensitive to how climatic and socio-economic criteria are weighted relative to each other, while the stable core identified in central Amman (by definition present in all scenarios) anchors this otherwise divergent prioritization landscape.

3.4. Discussion

Overall, our so-called 'temperate, need-based' approach yielded opposite results to a locally adapted 'semi-arid, feasibility-based' approach. Though this divergence was expected, it is critically important, as it highlights the urgent need to develop locally adapted prioritization approaches for GR implementation. Specifically, when priority is given to areas where GR are most needed under water-limited conditions, costly long-term irrigation and maintenance may be needed for plant survival, which may become limiting factors [47,48], resulting in priorities that may be difficult to realize from an economic point of view.

This mismatch is not simply a Global North versus Global South issue, because even within temperate regions, climate-change-related constraints (e.g., increasing drought frequency or heat stress) will couple needs and feasibility. Although recent multi-criteria assessments have advanced spatial greening prioritization using a need-based approach, they assumed equal feasibility across urban space [11,16]. Our findings, therefore, reinforce growing calls to move beyond *need* toward *feasibility* and a context-related prioritization approach using a combination of economic and ecological criteria.

It is commonly assumed that low-income areas spatially coincide with poor environmental conditions and should therefore be prioritized for GR implementation [49,50]. Interestingly, our results from Amman indicated that this is not necessarily the case. Historically, urban expansion and refugee influx shaped a socio-economic east–west divide in Amman [38,51]. However, in Western Amman, continued urban intensification appears to have restructured this pattern. Namely, wealthier residents are now primarily concentrated in the southwest (Abdoun corridor) and along the Airport Road [JFNC report; 32], which do not match tightly with the city's more climatically favourable zones in the northwest. This has created a new socio-economic northwest–southwest divide.

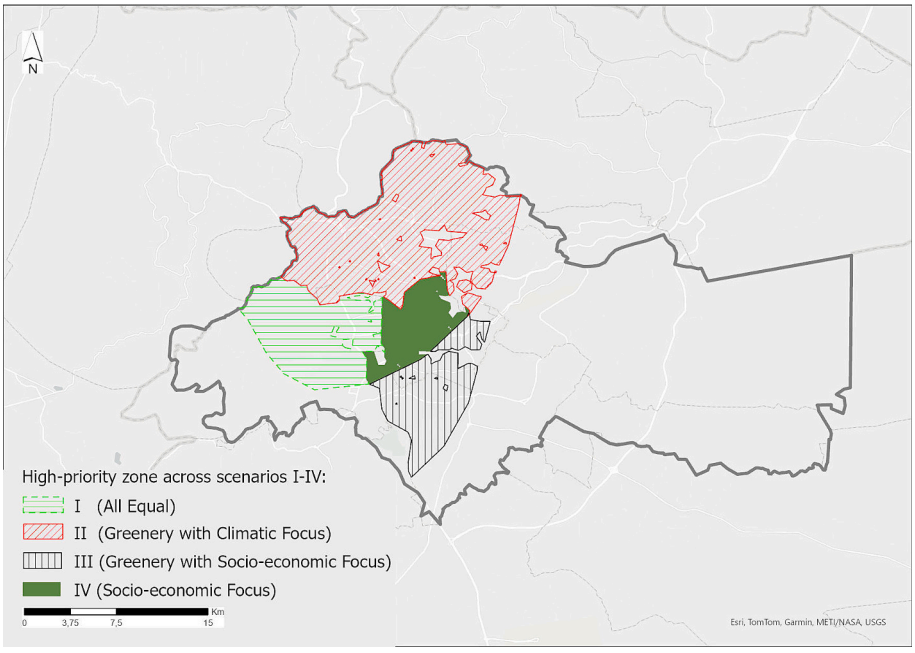


Fig. 6. Sensitivity of the high-priority zone across Scenarios I–IV. Western areas shifted along the north–south axis with changes in weighting scenarios, reflecting trade-off zones between the ecological–climatic (red), ecological–socio-economic (black), and equal-weight scenario (bright green). Core areas with high socio-economic weighting (dark green) remained stable across all scenarios. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Our scenario analyses revealed strong sensitivity to the weighting of climatic vs. socio-economic criteria. Increasing the weight of climatic suitability consistently shifted high-priority zones toward the north-western parts of Amman, where vegetation survival is more likely. In contrast, emphasizing socio-economic feasibility shifted priorities toward the south-west, where financial capacity is higher but climatic conditions are less favourable.

Previous studies sometimes suggest a one-size-fits-all solution without testing whether environmental and socio-economic criteria were spatially correlated [7,11,16]. When such criteria are strongly aligned, a single recommended prioritization may be adequate. However, when criteria are spatially decoupled, as in Amman, the produced high-priority zone reflects areas of overlap among competing objectives and masking areas that are highly sensitive to weighting changes, yet well suited for GR implementation under specific objectives. In our study, by comparing multiple weighting scenarios, we found substantial shifts in priority areas surrounding a persistent central overlap zone, revealing trade-offs between climatic and socio-economic criteria that would not have been captured in a single-scenario approach.

Despite the sensitivity of the suburbs to different weightings, a strikingly consistent high-priority core persisted in central Amman across all scenarios. The identification of such a 'low regret' area is crucial because it is robust against uncertainty in planning priorities. Similar efforts to identify spatial robustness are rare in GR studies, although Langemeyer et al. (2020) emphasized the importance of testing multiple scenarios to avoid misleading prioritization outcomes. This strengthens the transparency of the prioritization process and supports decision-making under competing objectives.

The weighted multi-criteria tool applied here [52] appears particularly relevant in contexts such as Amman, where climatic stress, water scarcity, and socio-economic constraints strongly interact, although it should not be considered universally applicable without further testing. Yet, one lesson learned here is that irrespective of context, prioritization tools should not prescribe a single optimal solution but rather reveal the full range of feasible options. This may reduce the potential risk of GR implementation failures [16,53] and ideally identify low-regret areas. This is also important because the tool does not make decisions autonomously but instead operationalizes explicitly defined criteria and weightings and translates them into a spatial representation of alternative outcomes [46,54]. This facilitates the visualization of flexible adjustment of weights and thus is an ideal format for planners to explore multiple future scenarios, aligned with available resources, and shifting policy objectives. Thus, the approach is a suitable decision-support tool.

Possible outcomes of our scenario analyses for practical management may be realized via different strategies. First, one may start with the central areas as low-regret entry points. Then, GR implementation could extend toward climatically favourable but economically less feasible areas, which may require some governmental support. Alternatively, GR implementation may extend toward socio-economically favourable but climatically constrained areas, depending on whether a careful roof design and water-efficient technologies may be implemented after an awareness campaign. Such stepwise strategies may provide a realistic GR implementation pathway that could support urban biodiversity but also microclimatic conditions under arid conditions.

It should be noted that prioritization of socio-economically disadvantaged zones occurred in Scenario II (Ecological/Climatic), which would require government support, where financial constraints can limit GR adoption [7]. Such a prioritization places greater responsibility on a government that may already be burdened by high debt, as in Amman, Jordan. On the other hand, our implementation strategy in Scenario III (Ecological/Socio-economic) is more affordability-oriented, as it prioritizes areas where GR are more likely to be implemented by wealthier individuals. This may reduce the financial burden on the government and may encourage private investment. However, relying on wealthier individuals for GR implementation is insufficient because wealthy individuals represent only a small fraction of the population, and when focusing on socio-economic affordability (Scenario IV), the suitable area shrank dramatically.

This study represents one of the first attempts to apply a locally adapted multi-criteria approach for prioritizing GR implementation under real-world conditions in a large, climatically and economically challenging city. Such approaches are increasingly used to support urban greening and climate adaptation planning, while also raising environmental justice considerations due to uneven benefit distribution [55]. Urban greening and blue–green infrastructure are widely recognized as effective nature-based solutions for improving urban sustainability, particularly for heat mitigation and ecosystem services [56]. GR are also increasingly adopted as a micro-scale nature-based solution for enhancing urban sustainability and resilience [57]. Similar GIS-based green space planning approaches have been applied in studies such as Al-Sareeh, Jordan, providing a useful basis for comparison, but overall, examples are still very rare. Compared to the GIS-based tool for optimal open green space allocation in Al-Sareeh, Jordan [58], our study differs in scope and objective. While Al-Sareeh focuses on optimizing ground-based green spaces in a small urban area (~1.2 km^2) using geology, soil, and rainfall, we developed a city-wide (~800 km^2) GR prioritization framework for the GAM, integrating ecological, climatic, and socio-economic layers. Where the single-scenario analysis of Al-Sareeh identified optimal locations for new green spaces and promoted equitable access, we focused on GR feasibility, biodiversity potential, and the transferability of temperate suitability logic to semi-arid conditions. Overall, our results indicate that unmodified temperate assumptions may produce unrealistic priority patterns in arid environments.

While our approach may provide a practical and scalable framework for prioritizing GR implementation city-wide, the results should be interpreted in light of data limitations. The analysis was constrained by subdistrict-level data, which limited the ability to distinguish fine-scale variation; for example, affluent neighbourhoods (such as Dabouq and Al-Hummar, around the royal palaces) in the north-western part of Amman could not be clearly distinguished. Higher-resolution datasets, as used in previous studies [11,12], could improve spatial precision; however, such data remain scarce in regions like Jordan, a common constraint in many Global South cities. As a result, the analysis may imply a level of spatial detail that is not fully supported by the input data. Additional uncertainty arises from map resolution, digitization, georeferencing, and overlay procedures, as well as from the use of proxy variables for biodiversity and socio-technical factors (e.g., potential GR performance under local climatic conditions, structural capacity, actual roof size, etc.). This suggests that the feasibility of GR implementation may be more complex than represented here. Therefore, our results should be viewed as a spatial prioritization framework under data-limited conditions rather than as a fully validated ecological, technical, or engineering assessment. Nevertheless, the approach provides a robust general overview and highlights the importance of generating and openly sharing locally derived datasets to support more accurate and context-sensitive greening strategies.

4. Conclusions

Our study suggests that GR planning in arid cities should be guided by feasibility rather than need, and that these two aspects may be inversely related. Specifically, temperate approaches that are not adapted to local conditions in drylands may fail to produce realistic spatial priorities in Global South cities. Therefore, a context-specific, scenario-based prioritization approach is essential for more realistic GR implementation, with relevance for climate, socio-economic adaptation, and biodiversity-oriented planning. Future work should further refine MCDA approaches through participatory weighting, long-term monitoring, and integration of low-cost irrigation and drought-resistant planting to improve feasibility under climate stress.

Declaration of ai-assisted technologies

During the preparation of this manuscript, the author used Grammarly and ChatGPT to improve language clarity, grammar, and spelling, but not to produce content. The authors personally reviewed, edited, and verified all AI-assisted content and assume full responsibility for the content and conclusions presented in the published article.

CRediT authorship contribution statement

Razan Alhmoud: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Andreas Braun:** Writing – review & editing, Validation, Resources. **Katja Tielbörger:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Razan Alhmoud reports financial support, administrative support, and travel were provided by the German Academic Exchange Service (DAAD) [grant number 57592434]. This study was conducted within SAGE Global Climate Centre, funded by the DAAD. All other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cacint.2026.100423.

Data availability

Data will be made available on request.

References

[1] Braaker S, Obrist MK, Ghazoul J, Moretti M. Habitat connectivity and local conditions shape taxonomic and functional diversity of arthropods on green roofs. J Anim Ecol 2017;86(3):521–31. https://doi.org/10.1111/1365-2656.12648.

[2] Ceballos G, Ehrlich PR, Raven PH. Vertebrates on the brink as indicators of biological annihilation and the sixth mass extinction. Proc Natl Acad Sci USA 2020; 117(24):13596–602. https://doi.org/10.1073/pnas.1922686117.

[3] M. F. Aronson *u. a.*, Biodiversity in the city: key challenges for urban green space management", Front Ecol Environ, 15, 4, 189–196, 2017, doi:https://doi.org/10.1002/fee.1480.

[4] WHO und The World Health Organization. Urban Green Spaces: A Brief for Action. 2017 [Online]. Available at: https://iris.who.int/server/api/core/bitstreams/ac6c76f4-7840-404f-8b8b-1ce3de666bc1/content.

[5] Cardinali M. Evaluating the Impact of Nature-based Solutions: A Summary for Policy Makers. 2021. 10.2777/521937.

[6] Shafique M, Kim R, Rafiq M. Green roof benefits, opportunities and challenges – a review. Renew Sustain Energy RevRenew Sustain Energy Rev 2018;90:757–73. https://doi.org/10.1016/j.rser.2018.04.006.

[7] Joshi MY, Selmi W, Binard M, Nys G-A, Teller J. Potential for urban greening with green roofs: a way towards smart cities. ISPRS Ann Photogramm Remote Sens Spat Inf SciISPRS Ann Photogramm Remote Sens Spat Inf Sci 2020;VI-4/W2–2020: 87–94. https://doi.org/10.5194/isprs-annals-VI-4-W2–2020-87-2020.

[8] Dromgold JR, Threlfall CG, Norton BA, Williams NSG. Green roof and ground-level invertebrate communities are similar and are driven by building height and landscape context. J Urban Ecol 2020;6(1):juz024. https://doi.org/10.1093/jue/juz024.

[9] Recher HF. The theory of island biogeography revisited. Austral Ecol 2011;36(7): e36–7. https://doi.org/10.1111/j.1442-9993.2010.02226.x.

[10] Silva CM, Flores-Colen I, Antunes M. Step-by-step approach to ranking green roof retrofit potential in urban areas: a case study of Lisbon, Portugal. Urban For Urban Green 2017;25:120–9. https://doi.org/10.1016/j.ufug.2017.04.018.

[11] Grunwald L, Heusinger J, Weber S. A GIS-based mapping methodology of urban green roof ecosystem services applied to a central European city. Urban For Urban Green 2017;22:54–63. https://doi.org/10.1016/j.ufug.2017.01.001.

[12] Velázquez J, Anza P, Gutiérrez J, Sánchez B, Hernando A, García-Abril A. Planning and selection of green roofs in large urban areas. Application to Madrid metropolitan area. uUrban gGreen Infrastruct – cConnect People Nat Sustain Cities 2019;40:323–34. https://doi.org/10.1016/j.ufug.2018.06.020.

[13] Gedge D, Kadas G. und. Green roofs and biodiversity, gehalten auf der world green roof congress Basel, Switzerland. 2005. p. 177.

[14] Pearce Huma, Walters Charlotte L. Do green roofs provide habitat for bats in urban areas? Acta Chiropterologica 2012;14(2):469–78. https://doi.org/10.3161/150811012X661774.

[15] Herrera-Gomez SS, Quevedo-Nolasco A, Pérez-Urrestarazu L. The role of green roofs in climate change mitigation. A case study in Seville (Spain). Build EnvironBuild Environ 2017;123:575–84. https://doi.org/10.1016/j.buildenv.2017.07.036.

[16] Langemeyer J, Wedgwood D, McPhearson T, Baró F, Madsen AL, Barton DN. Creating urban green infrastructure where it is needed – a spatial ecosystem service-based decision analysis of green roofs in Barcelona. Sci Total Environ 2020; 707:135487. https://doi.org/10.1016/j.scitotenv.2019.135487.

[17] Rubaszek J, Szymanowski M, Michalski A, Tatko R, Weber-Siwirska M. Procedure for the selection and evaluation of prefabricated housing buildings for the implementation of green roofs in the context of urban heat island mitigation. The example of Wrocław, Poland. PLoS One 2021;16:e0258641. https://doi.org/10.1371/journal.pone.0258641.

[18] Todeschini CC, Fett-Neto AG. Life at the top: extensive green roof plant species and their traits for urban use. Plants 2025;14(5):735. https://doi.org/10.3390/plants14050735.

[19] Vanuytrecht E, Van Mechelen C, Van Meerbeek K, Willems P, Hermy M, Raes D. Runoff and vegetation stress of green roofs under different climate change scenarios. Landsc Urban Plan 2014;122:68–77. https://doi.org/10.1016/j.landurbplan.2013.11.001.

[20] Nagase A, Dunnett N. Drought tolerance in different vegetation types for extensive green roofs: effects of watering and diversity. Landsc Urban Plan 2010;97(4): 318–27. https://doi.org/10.1016/j.landurbplan.2010.07.005.

[21] Raimondi A, Marchioni M, Sanfilippo U, Becciu G. Vegetation survival in green roofs without irrigation. Water 2021;13(2):136. https://doi.org/10.3390/w13020136.

[22] Benvenuti S, Bacci D. Initial agronomic performances of Mediterranean xerophytes in simulated dry green roofs. Urban EcosystUrban Ecosyst 2010;13(3):349–63. https://doi.org/10.1007/s11252-010-0124-9.

[23] Borràs JG, Lerma C, Mas Á, Vercher J, Gil E. Energy efficiency evaluation of green roofs as an intervention strategy in residential buildings in the field of Spanish climate. Buildings 2022;12(7):959. https://doi.org/10.3390/buildings12070959.

[24] Varela-Stasinopoulou DS, Nektarios PA, Tsanakas GF, Ntoulas N, Roukounakis GI, Economou AS. Impact of substrate depth and irrigation regime on growth, flowering and physiological indices of Greek sage (*Salvia fruticosa* mill.) grown on urban extensive green roof systems. Ecol Eng 2023;186:106816. https://doi.org/10.1016/j.ecoleng.2022.106816.

[25] Paço TA, Cruz de Carvalho R, Arsénio P, Martins D. Green roof design techniques to improve water use under Mediterranean conditions. Urban Sci 2019;3(1):14. https://doi.org/10.3390/urbansci3010014.

[26] Rayner JP, Farrell C, Raynor KJ, Murphy SM, Williams NSG. Plant establishment on a green roof under extreme hot and dry conditions: the importance of leaf succulence in plant selection. Urban For Urban Green 2016;15:6–14. https://doi.org/10.1016/j.ufug.2015.11.004.

[27] Szota C, Farrell C, Williams NSG, Arndt SK, Fletcher TD. Drought-avoiding plants with low water use can achieve high rainfall retention without jeopardising survival on green roofs. Sci Total Environ 2017;603–604:340–51. https://doi.org/10.1016/j.scitotenv.2017.06.061.

[28] Karimian Z, Mahdizadeh S. Microclimate effects and outdoor thermal comfort of green roof types in hot and dry climates: modelling in the historic city of Yazd, Iran. PLoS One 2025;20.

[29] Malpede M, Percoco M. Climate, desertification, and local human development: evidence from 1564 regions around the world. Ann Reg Sci 2023;72:1–29. https://doi.org/10.1007/s00168-022-01204-z.

[30] Tian Y, Jim CY. Factors influencing the spatial pattern of sky gardens in the compact city of Hong Kong. Landsc Urban Plan 2011;101(4):299–309. https://doi.org/10.1016/j.landurbplan.2011.02.035.

[31] Nektarios PA, Ntoulas N. und. Designing green roofs for arid and semi-arid climates. The route towards the adaptive approach, in *Acta Horticulturae*. Leuven, Belgium: International Society for Horticultural Science (ISHS); 2017. p. 197–202. 10.17660/ActaHortic.2017.1189.39.

[32] UNDP, The United Nations Development Programme. und. Jordan's Fourth National Communication on Climate Change (4NC). 2022 [Online]. Available at: https://www.undp.org/sites/g/files/zskgke326/files/2023-10/jordans_fourth_national_communication_report._spread_view_0.pdf.

[33] Khoury F, Eid E-M. State of Jordan's birds report. Amman, Jordan: The Royal Society for the conservation of nature; 2013 [Online]. Available at: https://www.researchgate.net/profile/Ehab-Eid/publication/292157922_The_State_of_Jordan's_Birds_Report/links/56afa2b608ae9f0ff7b26dfe/The-State-of-Jordans-Birds-Report.pdf.

[34] AECOM. Amman Green City Action Plan, Greater Amman Municipality and European Bank for Reconstruction and Development (EBRD), Amman, Jordan. 2021 [Online]. Available at: https://www.amman.jo/site_doc/AmmanGreen2021.pdf.

[35] UN-Habitat. Amman Spatial Profile, Greater Amman Municipality and Swiss State Secretariat for Economic Affairs (SECO). 2022 [Online]. Available at: https://unhabitat.org/sites/default/files/2022/04/220411-final_amman_profile.pdf.

[36] Al-Bilbisi H. Spatial monitoring of urban expansion using satellite remote sensing images: a case study of Amman City, Jordan. Sustainability 2019;11(8):2260. https://doi.org/10.3390/su11082260.
[37] Bazian. GAM cites challenges to expanding parks as WHO standards reveal 'great shortage' of green areas, The Jordan Times. 2019 [Online]. Available at: https://jordantimes.com/news/local/gam-cites-challenges-expanding-parks-who-standards-reveal-'great-shortage'-green-areas.
[38] Alkaraki KF, Hazaymeh K, Al-Tarawneh OM, Jawarneh RN. Deep learning-based modeling of land use/land cover changes impact on land surface temperature in greater Amman municipality, Jordan (1980–2030). GeoJournal 2024;89(4):170. https://doi.org/10.1007/s10708-024-11187-3.
[39] GTC, Greening the Camps". 2026 [Online]. Available at: https://www.facebook.com/GreeningTheCamps/ (Accessed 12/2024).
[40] DA, Durrat Almanal for Development and Training". 2026 [Online]. Available at: https://www.facebook.com/durratalmanal.org.jo/about (Accessed 12/2024).
[41] CSBE. Urban Agriculture in Amman: A Holistic View. 2019 [Online]. Available at: https://static1.squarespace.com/static/5671433fc647ad9f55531f40/t/5d9edca97d71f81ac857e9c3/1570692337893/Urban+Agriculture+Report-EN.pdf.
[42] ESA. WorldCover 10m. The European Space Agency; 2021 [Online]. Available at: https://earthmap.org/?aoi=jo&boundary=level0&feature&features&layers=%7B"WorldCover10m"%3A%7B"opacity"%3A1%7D%7D&map=%7B"center"%3A%7B"lat"%3A31.302530241690288%2C"lng"%3A37.129107364108556%7D%2C"zoom"%3A7%2C"mapType"%3A"roadmap"%7D&statisticsOpen=true.
[43] GitHub. GlobalMLBuildingFootprintsTest. 2022 [Online]. Available at: https://github.com/ao68/GlobalMLBuildingFootprintsTest.
[44] Esri, ArcGIS Pro. Environmental Systems Research Institute (ESRI), Redlands, CA. 2022.
[45] Hussain RI, Frank T, Kratschmer S. More insect species are supported by green roofs near public gardens. J Insect Conserv 2023;27(6):941–6. https://doi.org/10.1007/s10841-023-00510-x.
[46] Getter KL, Rowe DB. The role of extensive green roofs in sustainable development. HortScience 2006;41(5):1276–85. https://doi.org/10.21273/HORTSCI.41.5.1276.
[47] Speak AF, Rothwell JJ, Lindley SJ, Smith CL. Rainwater runoff retention on an aged intensive green roof. Sci Total Environ 2013;461–462:28–38. https://doi.org/10.1016/j.scitotenv.2013.04.085.
[48] Venter ZS, Figari H, Krange O, Gundersen V. Environmental justice in a very green city: spatial inequality in exposure to urban nature, air pollution and heat in Oslo, Norway. Sci Total Environ 2023;858:160193. https://doi.org/10.1016/j.scitotenv.2022.160193.
[49] Venter ZS, Barton DN, Martinez-Izquierdo L, Langemeyer J, Baró F, McPhearson T. Interactive spatial planning of urban green infrastructure – retrofitting green roofs where ecosystem services are most needed in Oslo. Ecosyst Serv 2021;50:101314. https://doi.org/10.1016/j.ecoser.2021.101314.
[50] Khirfan L, Mohtat N, Daub B. Reading an urban palimpsest: how the gradual loss of an urban stream impacts urban form's connections and ecosystem functions. Front Water 2021;3(0):754679. https://doi.org/10.3389/frwa.2021.754679.
[51] Şenol Hİ, Yiğit AY, Ulvi A. GIS-based multi-criteria analysis for urban afforestation planning in semi-arid cities. Forests 2025;16(7):1064. https://doi.org/10.3390/f16071064.
[52] Geneletti D. Combining stakeholder analysis and spatial multicriteria evaluation to select and rank inert landfill sites. Waste ManagWaste Manag 2010;30(2):328–37. https://doi.org/10.1016/j.wasman.2009.09.039.
[53] Malczewski J. GIS-based multicriteria decision analysis: a survey of the literature. Int J Geogr Inf Sci 2006;20(7):703–26. https://doi.org/10.1080/13658810600661508.
[54] Malczewski J, Rinner C. Multicriteria Decision Analysis in Geographic Information Science1. Springer; 2015. https://doi.org/10.1007/978-3-540-74757-4.
[55] Gupta A, De B. Enhancing the city-level thermal environment through the strategic utilization of urban green spaces employing geospatial techniques. Int J BiometeorolInt J Biometeorol 2024;68, 10:2083–101. https://doi.org/10.1007/s00484-024-02733-2.
[56] Gupta A, De B. A systematic review on urban blue-green infrastructure in the south Asian region: recent advancements, applications, and challenges. Water Sci Technol 2024;89(2):382–403. https://doi.org/10.2166/wst.2024.014.
[57] G. Mihalakakou *u. a.*, Green roofs as a nature-based solution for improving urban sustainability: progress and perspectives", Renew Sust Energ Rev, 180, S. 113306, Juli 2023, doi:https://doi.org/10.1016/j.rser.2023.113306.
[58] Banisalman AM, Elsharkawy MM, Labin AE. Tool for the establishment of optimal open green spaces using GIS and nature-based solutions: Al-Sareeh (Jordan) case study. Sustainability 2025;17(19):8647. https://doi.org/10.3390/su17198647.